

Chapter 14 at a glance: e-values vs. p-values

	e-value	p-value
Monotone role of X	E increasing in X	P decreasing in X
Natural bound (Prop. 14.1)	$E \leq 1/S(X; \mathbb{P})$	$P \geq S(X; \mathbb{P})$
Duality (Thm. 14.2)	$\sup_{E \in \mathcal{E}^X} E$	$= 1 / \inf_{P \in \mathcal{U}^X} P$
Structural representation	$E \leq E^{\mathbb{P}}[dQ/d\mathbb{P} \mid X]$ (Thm. 14.4)	$P \geq \mathbb{P}(T(X') \leq T(X) \mid X)$ (Thm. 14.5)
What it "is"	a conditional expectation of a likelihood ratio	a conditional probability of a rarer outcome
Existence of exact, powered version (Thm. 14.6)	$\text{Conv}(\mathcal{Q}) \cap \text{Span}(\mathcal{P}) = \emptyset$	$\text{Conv}(\mathcal{Q}) \cap \text{Span}(\mathcal{P}) = \emptyset$ (same condition, via duality)